

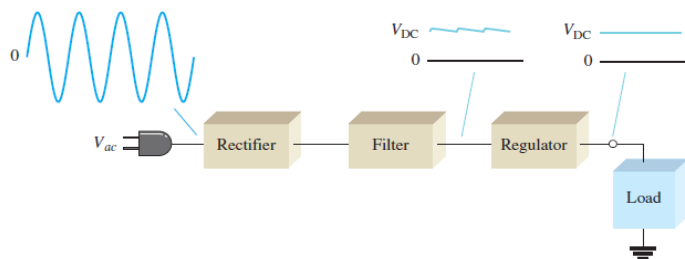


Experiment Name

Design power supply

Objectives

In this Lecture you to be able to create simple power supply using the design created by the group. A power supply is an electronic device that supplies electric energy to an electrical load. Its main function is to convert one form of electrical energy to another energy. Power supplies will sometimes employ a transformer to convert the input voltage to a higher or lower AC voltage. A rectifier is used to convert the transformer output voltage to a varying DC voltage, which in turn is passed through an filter to convert it to an unregulated DC voltage.

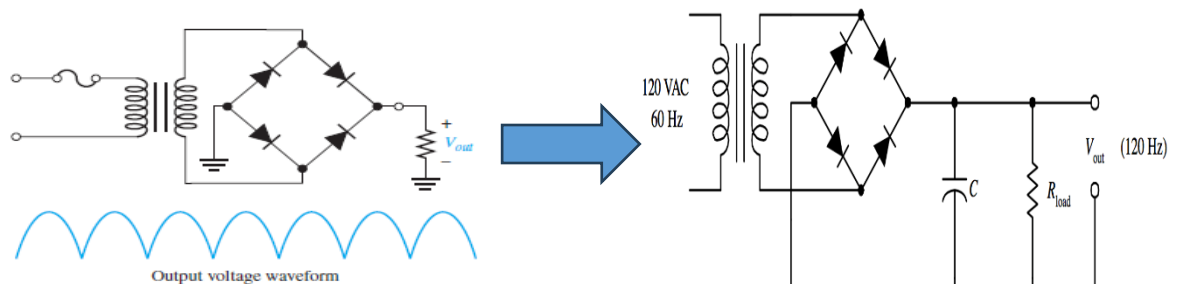


1. design un regulated power supply

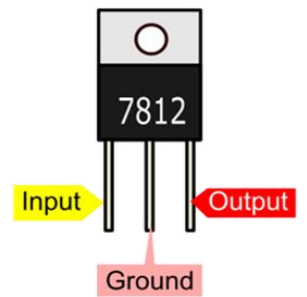
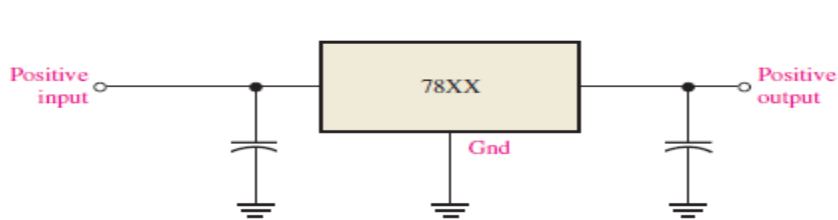
1. choose $R = 1k\Omega$ and the output voltage at the secondary transformer is 20 VP-P ,60 HZ.

$$V_{p(out)} = V_{p(sec)} - 1.4 V$$

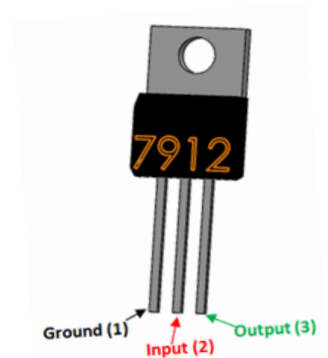
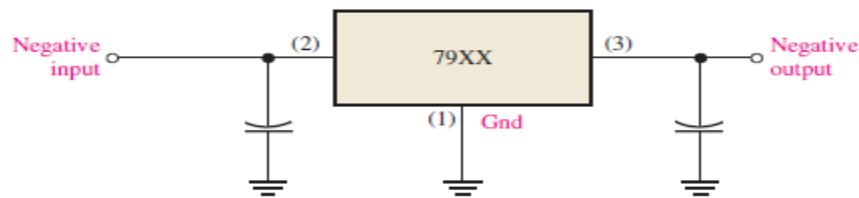
2. Measure the unfiltered peak full-wave rectified voltage $v_p(out) = \underline{\hspace{2cm}}$
3. Connect a capacitor input voltage with a (10uf and measure output voltage of a filtered rectifier for value of capacitance $V_{r(pp)} \cong \left(\frac{1}{fR_LC}\right)V_{p(rect)} = \underline{\hspace{2cm}}$
4. What happens to the output voltage of a rectifier if the filter capacitor change from 10uf to 100uf ? $\underline{\hspace{2cm}}$



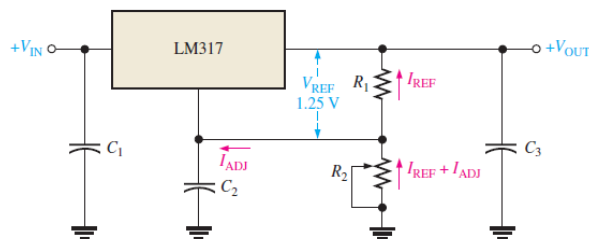
2. with regulated power supply (Fixed Positive Linear Voltage Regulators)



Fixed Negative Linear Voltage Regulators



Adjustable Positive Linear Voltage Regulator

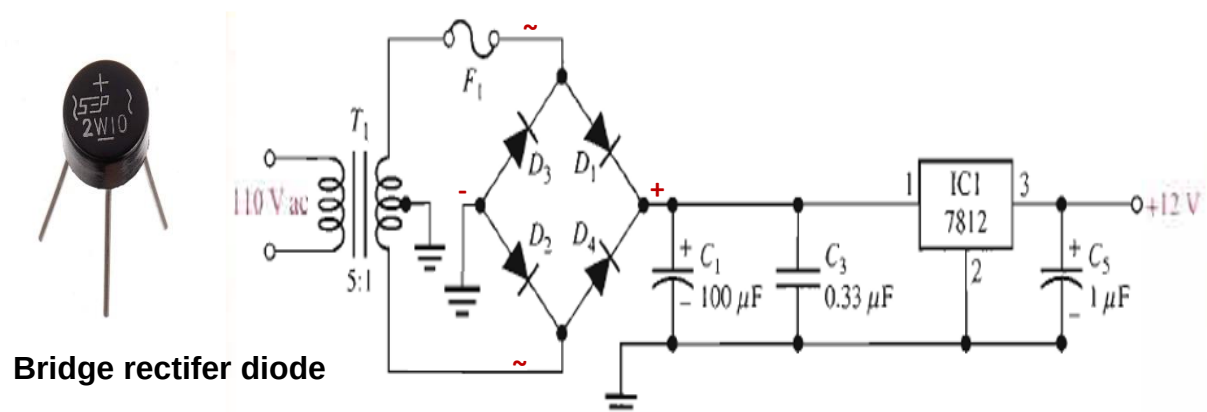


$$V_{OUT} = V_{REF} \left(1 + \frac{R_2}{R_1} \right) + I_{ADJ} R_2$$

$$V_{Out} = 1.25 \left(1 + \frac{R_2}{R_1} \right)$$

Note * Adjustable current (I_{ADJ}) is Very less and can be neglected, output voltages lie in the range 3-40 V

2. design regulated power supply



Bridge rectifier diode

2 page or 3

Note* The 7812 IC is a commonly used linear regulator. Input voltage can range from 12 – 35VDC and it outputs a fixed 12V at over 1A of current and up to 2.2A of surge current

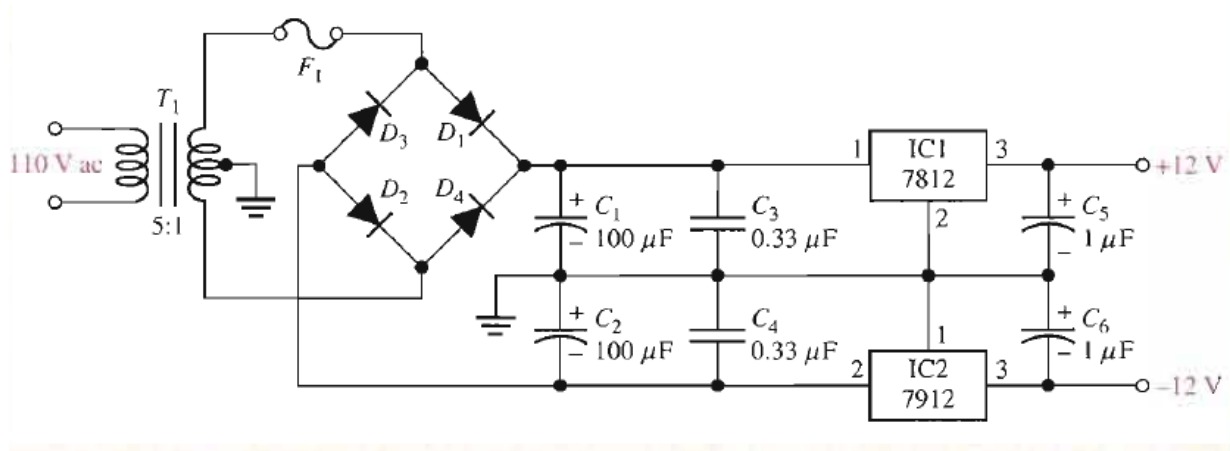
● 3. Design the regulated power supply to provide dual polarities dc voltages of $\pm 12V$. the output voltage at the secondary transformer 20 vpp 60HZ.

(a) Based on the dc current amount of dc c required by the receiver do the regulators needs to attached to heated sink ?

(b) why should be connected 0.33uf and 1uf capacitor to the power supply circuit

(c) why connected a small-value capacitor 0.33uf in parallel with a large capacitance 100uf

● 3. Modification the regulated power supply to provide dual polarities dc voltages of $\pm 12V$



● 4. Recall the 7812IC provided a 12 V regulated output .develop the circuit and Write report on the dc variable power supply using 7809 to provide an output voltage from 9v to 30v an organized format that includes the following

(a) Physical description of the circuit

(b) A discussion of the operation of the circuit

(c) A list of the specification